

Uncertainty-Gated Test-Time Adaptation for Reliable Cross-Hospital Histopathology Classification

UTTA-Med | Camelyon17-WILDS | ResNet-18 | 5 seeds {42, 123, 2024, 7, 99}
Numbers frozen. Target hospital labels used only for final evaluation.

Abstract

Models trained on histopathology patches from source hospitals degrade on an unseen hospital. Test-time adaptation (TTA) can recover some of that drop, but entropy minimization on every target sample can also amplify errors. We restrict TTA to low-uncertainty samples under real Camelyon17-WILDS shift (source hospitals {0,3,4} to target hospital 2). We compare source-only, Tent, EATA, and four gates (random, softmax confidence, predictive entropy, MC-Dropout variance) under a frozen protocol: BatchNorm affine updates only, running statistics frozen, TTA learning rate 1e-5, and gate thresholds chosen on hospital 1.

On hospital 2, source AUROC is 0.930 +/- 0.009. Fair Tent is unstable: it collapses on 2/5 seeds (mean AUROC 0.698 +/- 0.324, harm 0.187 +/- 0.179). Selective gates never collapse, raise F1 on all five seeds, and cut harm to 0.007 +/- 0.003. UTTA-Med (MC-Dropout) does not outperform softmax confidence (delta AUROC < 0.001 on every seed). Random gating also collapses on 2/5 seeds, so the benefit is which samples are used, not merely using fewer of them.

Takeaway. Gating is what makes TTA reliable; MC-Dropout is a valid error signal but is redundant with confidence. The contribution is a leakage-controlled reliability profile, not a claim that Bayesian uncertainty uniquely wins.

1. Introduction

A classifier trained on labeled patches from source hospitals is often deployed on a new scanner, stain, and patient population. TTA updates a frozen source checkpoint using unlabeled target images, typically by entropy minimization on BatchNorm affine parameters (Tent). The failure mode is harmful adaptation: a wrong prediction becomes an update signal and the model degrades.

UTTA-Med uses a hard gate: adapt only if MC-Dropout predictive variance $U(x) < \tau$, otherwise skip. We do not claim the first uncertainty-aware TTA method. Closest prior work: EATA (entropy + diversity) and SAR (sharpness-aware TTA). Our difference: (i) an explicit MC-Dropout variance gate, (ii) matched-coverage controls against random, confidence, and entropy, (iii) joint reporting of coverage, harm rate, and calibration on a leakage-controlled Camelyon17-WILDS split.

2. Method (frozen)

Splits (verified from the official loader, not from documents): train hospitals 0/3/4 (302,436 patches, 30 WSI); id-val 0/3/4 (33,560); OOD val hospital 1 (34,904, 10 WSI) for τ / lr only; test hospital 2 (85,054, 10 WSI). Train intersect OOD-val/test slides is empty.

Source: ResNet-18, 1 logit, BCE-with-logits, Dropout 0.5, Adam 1e-4, save-best on hospital-1 AUROC. TTA: continual one pass, BN gamma/beta only, running stats frozen, $\text{lr}=1\text{e-}5$. Empty gate: skip backward and step. τ = 70th percentile of hospital-1 U (rule frozen; numeric τ re-estimated per seed). $N_{\text{MC}} = 20$.

3. Results

Table 1. Hospital 2, mean +/- std, n = 5 seeds

Method	AUROC	F1	ECE	Harm	Cov.
Source-only	0.930 +/- 0.009	0.807 +/- 0.025	0.128 +/- 0.026	--	--
Tent	0.698 +/- 0.324	0.506 +/- 0.411	0.284 +/- 0.179	0.187 +/- 0.179	1.00
EATA	0.921 +/- 0.026	0.809 +/- 0.031	0.152 +/- 0.024	0.025 +/- 0.015	0.89
Random gate	0.697 +/- 0.304	0.487 +/- 0.385	0.298 +/- 0.166	0.194 +/- 0.170	0.70
Confidence	0.936 +/- 0.022	0.831 +/- 0.032	0.130 +/- 0.030	0.007 +/- 0.003	0.58
UTTA-Med	0.935 +/- 0.022	0.830 +/- 0.032	0.131 +/- 0.031	0.007 +/- 0.003	0.58

Entropy equals confidence in binary classification (omitted as a duplicate row). Tent/random collapse on seeds 7 and 99.

Per-seed AUROC (hospital 2)

Method	42	123	2024	7	99
Source	0.935	0.935	0.932	0.913	0.933
Tent	0.929	0.941	0.933	0.343	0.344
EATA	0.936	0.940	0.932	0.877	0.920
Random	0.922	0.937	0.897	0.354	0.373
Confidence	0.951	0.942	0.956	0.902	0.927
UTTA-Med	0.951	0.942	0.955	0.902	0.928

RQ1: shift is real (AUROC drop 0.070 +/- 0.009; F1 drop ~0.18). Sensitivity is the failure mode. RQ3: U vs error r = 0.40 +/- 0.02 on hospital 1; decile 1 error ~0%, decile 10 ~35-42%. RQ2: Tent collapses on 2/5 seeds even at lr=1e-5 with frozen BN stats. EATA never collapsed but does not beat source AUROC. RQ4: gating cuts harm ~25x; random at 70% coverage still collapses -- the mechanism is which samples, not fewer samples. RQ5: gated F1 improves on 5/5 seeds; AUROC mixed (up on 3/5). ECE is not a win vs source. RQ6: UTTA-Med ~ confidence (delta < 0.001 every seed).

4. Figures

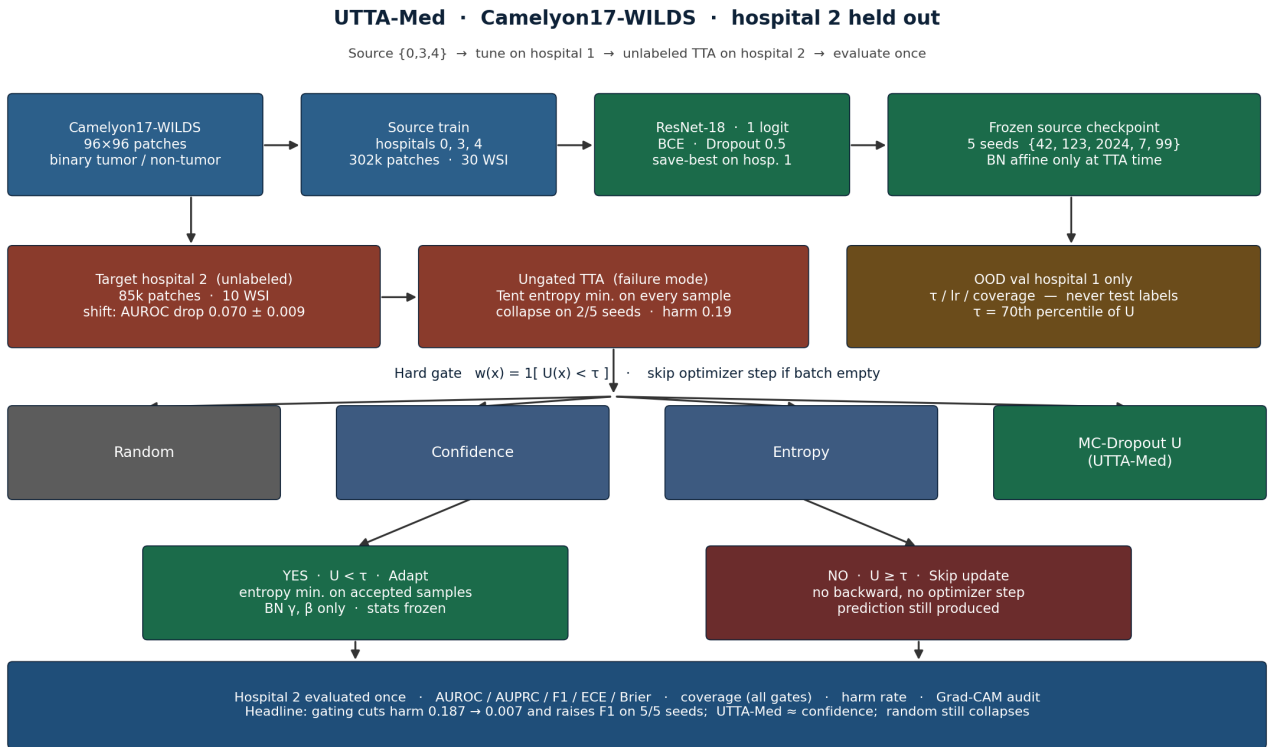


Figure 1. UTTA-Med pipeline. Target labels never enter tau, lr, or model selection.

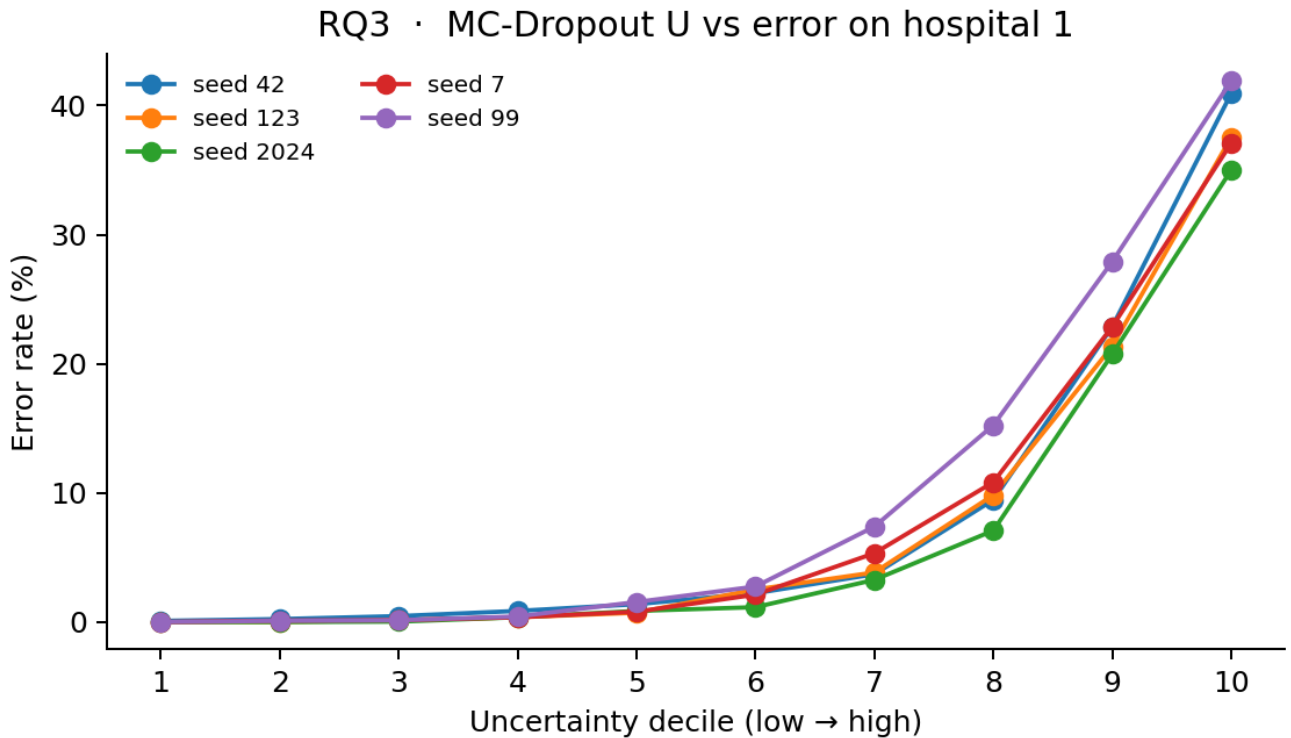


Figure 3. Uncertainty vs error on hospital 1, all five seeds. Mandatory gate before using U for TTA.

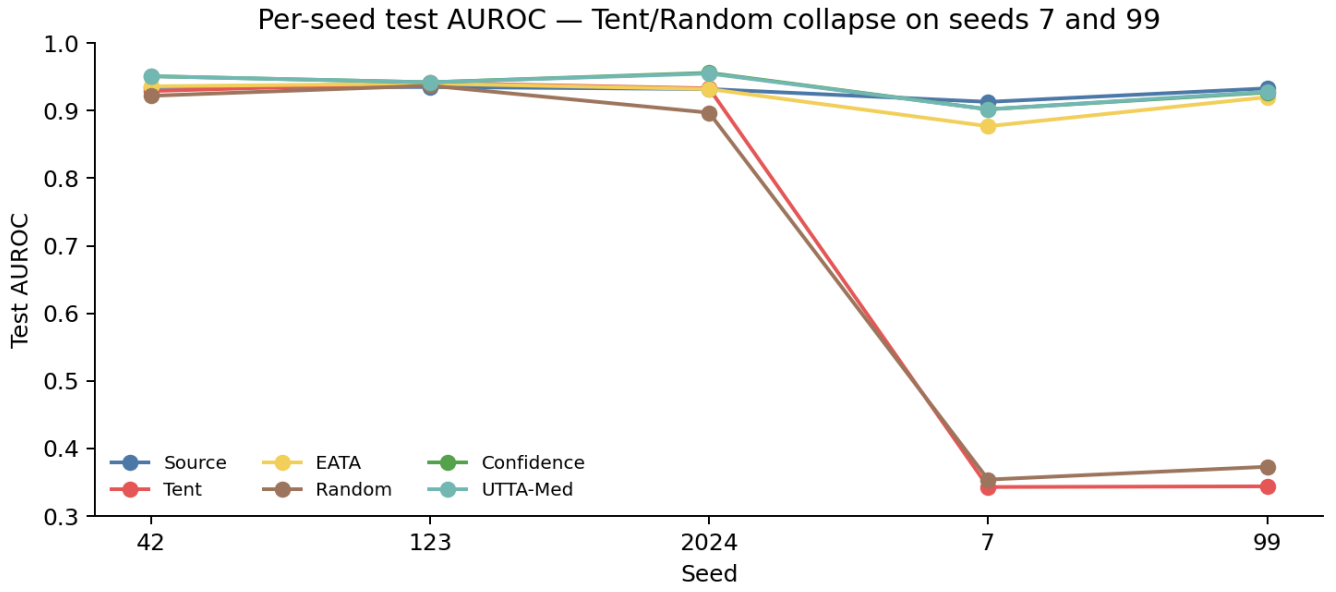


Figure 5. Per-seed hospital-2 AUROC. Collapse of Tent/random on seeds 7 and 99 is the reliability result.

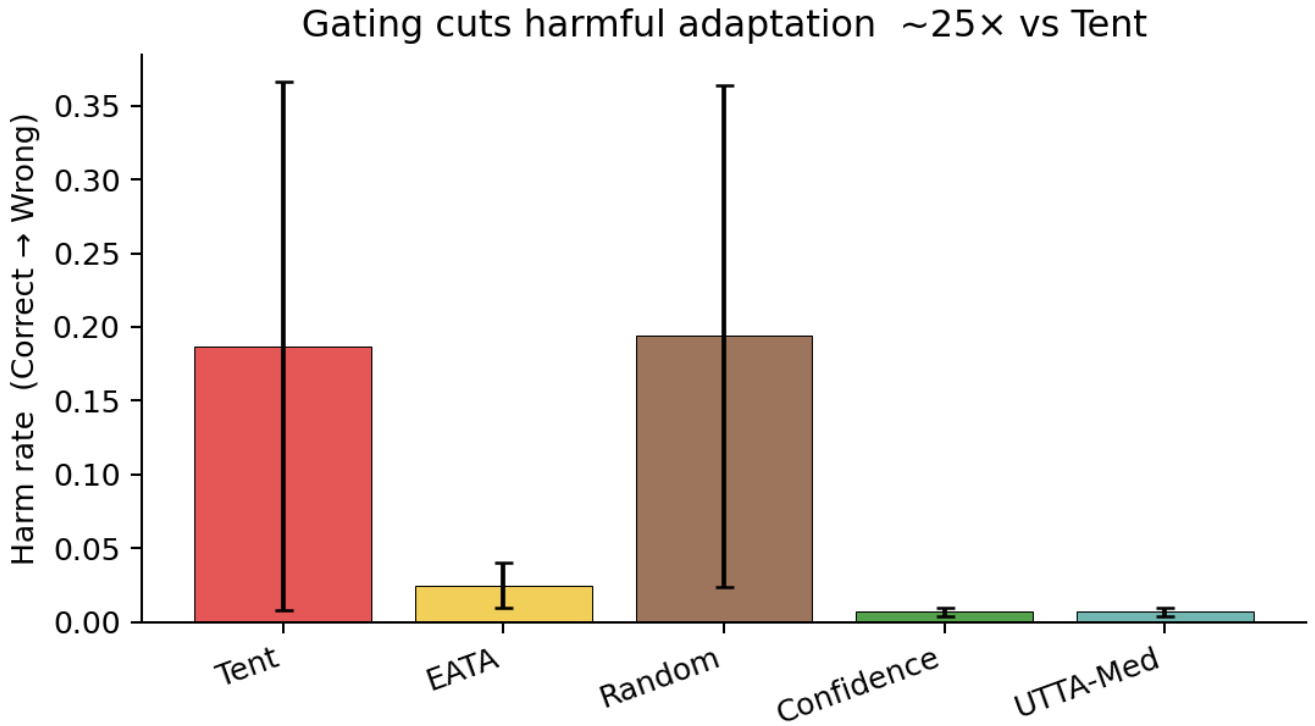


Figure 8. Harm rate. Gating prevents the Correct-to-Wrong flips that destroy Tent.

5. Grad-CAM audit (RQ7)

Seed-42 source model, 512 shuffled hospital-2 patches (TP 215, TN 226, FP 15, FN 56). Interpretability audit, not clinical validation. 96x96 maps are coarse. Blank CAMs on confident normals are a 1-logit Grad-CAM artifact.

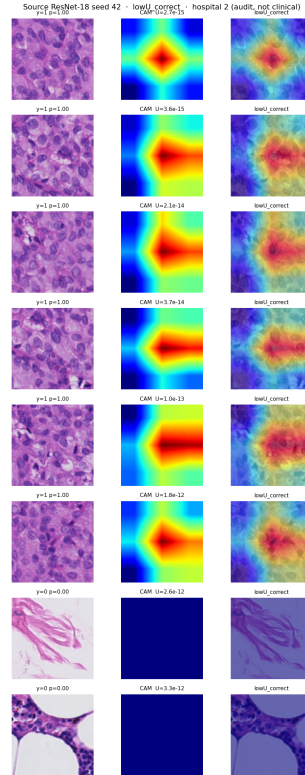


Figure 7a. Low-uncertainty correct: CAM on dense nuclei for tumors; fat/stroma often blank.

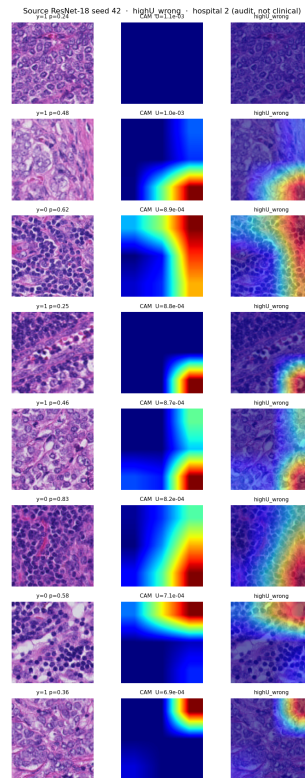


Figure 7b. High-uncertainty errors (p near 0.5, edge CAMs) -- the population the gate skips.

Residual failure mode: confident false positives on lymphocyte-dense tissue ($p > 0.96$) and confident false negatives ($p \sim 0.01$) still pass a hard gate.

6. Discussion and limitations

Positive statement: unrestricted entropy-minimization TTA is unsafe under real hospital shift; restricting updates to high-confidence / low-variance samples makes it safe and improves F1. Honest qualifier: MC-Dropout variance did not beat softmax confidence for this binary ResNet-18.

Limitations: (i) one dataset, one backbone, binary labels; (ii) seed-level CIs, not yet WSI-level bootstrap (10 test slides); (iii) Grad-CAM is source-only; (iv) test coverage of learned gates is below the val-matched 0.70; (v) Tent lr was selected on hospital 1 after larger lrs collapsed -- not Tent's original ImageNet hyperparameter; (vi) confident errors still pass a hard gate.

7. What not to claim

Do not write: first uncertainty-aware TTA for medical images; UTTA-Med outperforms confidence; Tent is useless without showing the 3 non-collapsed seeds; clinical validity of Grad-CAM; drop seeds 7 and 99.

Reproducibility

Data: HuggingFace wltjr1007/Camelyon17-WILDS parquet, filter by center. Checkpoints: camelyon17_resnet18_source_s{seed}_BEST.pt. Protocol: bn_freeze_stats, TTA lr 1e-5, N_MC=20, tau = val 70th percentile of U. Every headline number traces to camelyon17_resnet18_full_s{seed}.json. IEEE LaTeX: paper/ieee/utta_med.tex (Overleaf).